AI-GENERATED ENGINEERING PROTOTYPE - V2 - HUMAN REVIEW REQUIRED

Design basis: examples/sic-buck-400v-200v-5kw/design/design_decisions.json (schema 3, 2026-10-10) - single source of truth; electrical model automation/kicad_tools/sic_buck_power_model.py; generator automation/kicad_tools/generate_sic_schematics.py.
400 V (360-440 V) to 200 V / 25 A / 5 kW / 50 kHz synchronous buck; Wolfspeed C4MS025120K; TMS320F28335 control board on a stacked SELV mezzanine.
Net classes: HV_VIN, HV_SW, HV_PGND, HV_VOUT, DRV_HS, DRV_LS, DESAT_HS, DESAT_LS, SENSE_ISO - HV classes 3 mm, driver / DESAT / sensing islands 0.3 mm intra-class, PE 5 mm, CTRL_SELV and Default 0.25 mm (70 um copper).
Board rules (.kicad_dru): HV side to CTRL_SELV 9 mm on all layers; HV to PE 5 mm; HV functional 3 mm outer / 1 mm inner; HV to board edge 1 mm; 36 pin-field exceptions derived from pad geometry (9 barrier, 19 HV, 8 package pitch) - creepage review.
Barrier parts (UCC21710 x2, UCC14141 x3, AMC1311 x2, TMCS1133) need the 1.2 mm non-plated isolation slot of the baseline; the custom AIPE footprints embed it, the UCC21710 library footprint does not (layout task).
TO-247-4 pad order 1 D, 2 S, 3 KS, 4 G is an ASSUMPTION (Wolfspeed PRD 05966 front view); unverified parts and placeholders are listed in components/bom_power.csv notes and in the model docstring.
Evidence: native ERC manifests under verification/erc/<run>/run.json; review PDF under review/; this drawing has no CAD evidence beyond those runs. AI-GENERATED ENGINEERING PROTOTYPE - V2 - HUMAN REVIEW REQUIRED.

Input, half bridge, inductor, output, bleeders

文件: 01_power_stage.kicad_sch

Input, half bridge, inductor, output, bleeders (2)

文件: 01_power_stage_2.kicad_sch

High-side isolated gate driver and bias

文件: 02_gate_driver_hs.kicad_sch

High-side isolated gate driver and bias (2)

文件: 02_gate_driver_hs_2.kicad_sch

Low-side isolated gate driver and bias

文件: 03_gate_driver_ls.kicad_sch

Low-side isolated gate driver and bias (2)

文件: 03_gate_driver_ls_2.kicad_sch

Isolated voltage sensing island and its bias module

文件: 04_sensing_island.kicad_sch

Isolated voltage sensing island and its bias module (2)

文件: 04_sensing_island_2.kicad_sch

SELV strip: mezzanine, conditioning, relay and fan interface

文件: 05_selv_strip.kicad_sch

SELV strip: mezzanine, conditioning, relay and fan interface (2)

文件: 05_selv_strip_2.kicad_sch

Heatsink envelope, terminals, mounting, test points

文件: 06_mechanical.kicad_sch

AI-GENERATED ENGINEERING PROTOTYPE - V2 - HUMAN REVIEW REQUIRED

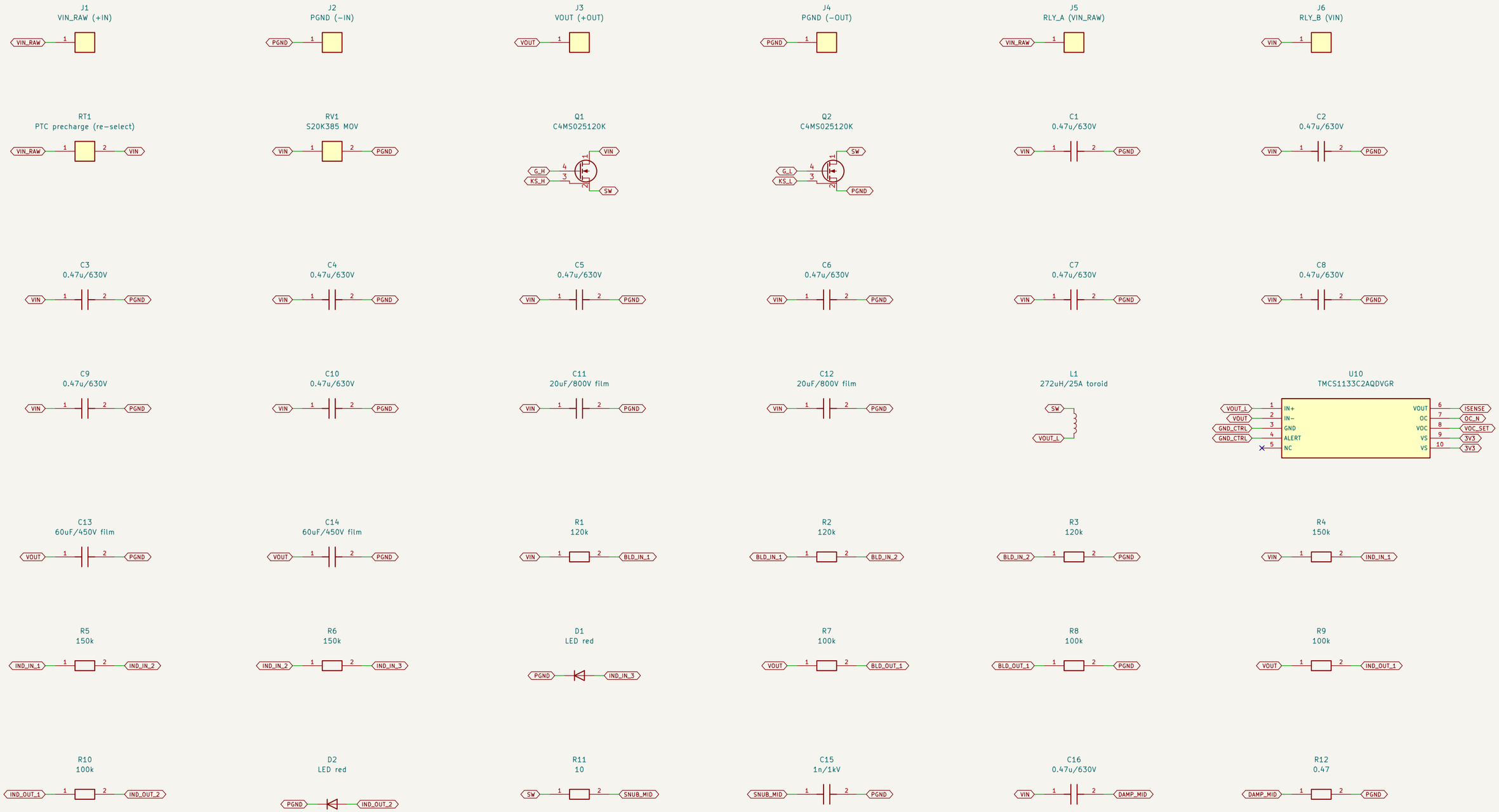
Sheet: /
File: AIPE_SiC_Buck_Power.kicad_sch

Title: AIPE V2 SiC buck power board - 400 V (360-440 V) to 200 V / 25 A

Size: A3	Date:	**Rev: V2**
KiCad E.D.A. 10.0.7		Id: 1/12

Input, half bridge, inductor, output, bleeders

Q1/Q2 pads 1 D, 2 S, 3 KS, 4 G follow the Wolfspeed PRD 05966 front-view lead order — ASSUMPTION, verify the footprint pad-1 side before fabrication. HV classes: VIN_RAW/VIN (HV_VIN), SW (HV_SW), VOUT_L/VOUT (HV_VOUT), PGND (HV_PGND) plus bleeder, indicator, ladder-tap, snubber and damping nodes. DNP: snubber, damping leg, MOV, PGND–PE link, Y2 capacitors.



AI-GENERATED ENGINEERING PROTOTYPE — V2 — HUMAN REVIEW REQUIRED

Sheet: /Input, half bridge, inductor, output, bleeders/
File: 01_power_stage.kicad_sch

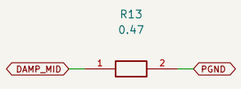
Title: Input, half bridge, inductor, output, bleeders

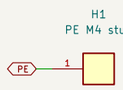
Size: A3	Date:	**Rev: V2**
KiCad E.D.A. 10.0.7		Id: 2/12

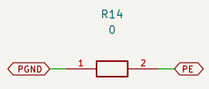
Input, half bridge, inductor, output, bleeders (2)

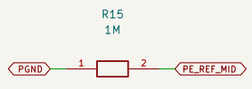
Q1/Q2 pads 1 D, 2 S, 3 KS, 4 G follow the Wolfspeed PRD 05966 front-view lead order — ASSUMPTION, verify the footprint pad-1 side before fabrication. HV classes: VIN_RAW/VIN (HV_VIN), SW (HV_SW), VOUT_L/VOUT (HV_VOUT), PGND (HV_PGND) plus bleeder, indicator, ladder-tap, snubber and damping nodes. DNP: snubber, damping leg, MOV, PGND–PE link, Y2 capacitors.

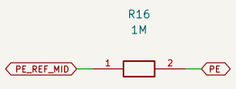




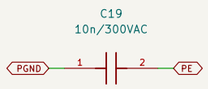












AI-GENERATED ENGINEERING PROTOTYPE — V2 — HUMAN REVIEW REQUIRED

Sheet: /Input, half bridge, inductor, output, bleeders (2)/
File: 01_power_stage_2.kicad_sch

Title: Input, half bridge, inductor, output, bleeders (2)

Size: A3	Date:	**Rev: V2**
KiCad E.D.A. 10.0.7		Id: 3/12

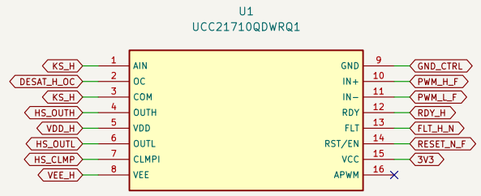
High-side isolated gate driver and bias

UCC21710 IN+ = PWM_H_F, IN- = PWM_L_F (hardware interlock KR-GD-009); AIN tied to COM (CR-01); APWM not connected. UCC14141 VDD-VEE 19.0 V (107 k / 16.2 k) and COM-VEE 4.0 V (9.76 k / 16.2 k) give +15 V / -4 V at the gate; RLIM 1 k is a placeholder (human review). Decoupling sits between the driver pins and the gate/Kelvin connection (KR-PCB-002); KS_H is never joined to SW on the PCB (KR-PCB-003).

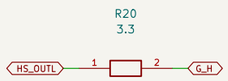


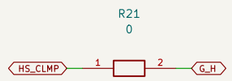








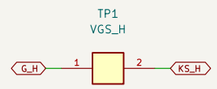






























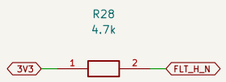


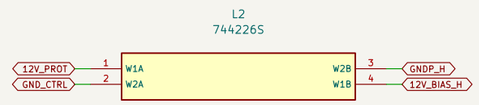




















AI-GENERATED ENGINEERING PROTOTYPE – V2 – HUMAN REVIEW REQUIRED

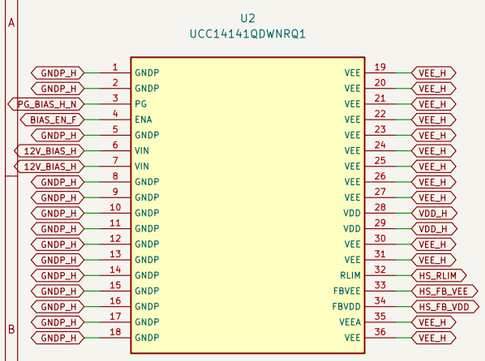
Sheet: /High-side isolated gate driver and bias/
File: 02_gate_driver_hs.kicad_sch

Title: High-side isolated gate driver and bias

Size: A3	Date:	**Rev: V2**
KiCad E.D.A. 10.0.7		Id: 4/12

High-side isolated gate driver and bias (2)

UCC21710 IN+ = PWM_H_F, IN- = PWM_L_F (hardware interlock KR-GD-009); AIN tied to COM (CR-01); APWM not connected. UCC14141 VDD-VEE 19.0 V (107 k / 16.2 k) and COM-VEE 4.0 V (9.76 k / 16.2 k) give +15 V / -4 V at the gate; RLIM 1 k is a placeholder (human review). Decoupling sits between the driver pins and the gate/Kelvin connection (KR-PCB-002); KS_H is never joined to SW on the PCB (KR-PCB-003).



























AI-GENERATED ENGINEERING PROTOTYPE — V2 — HUMAN REVIEW REQUIRED

Sheet: /High-side isolated gate driver and bias (2)/
File: 02_gate_driver_hs_2.kicad_sch

Title: High-side isolated gate driver and bias (2)

Size: A3	Date:	**Rev: V2**
KiCad E.D.A. 10.0.7		Id: 5/12

Low-side isolated gate driver and bias

Mirror of the high-side channel: IN+ = PWM_L_F, IN- = PWM_H_F; DESAT diode cathode on SW; KS_L is never joined to PGND on the PCB (KR-PCB-003). Isolation slot under the library SOIC-16W footprint is a layout task.

R35
100



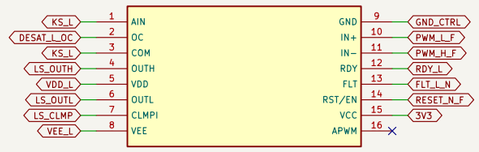
C40
100p/50V



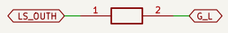
R36
10k



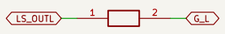
U3
UCC21710QDWRQ1



R37
10



R38
3.3



R39
0



R40
10k



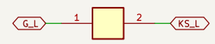
R41
1



C41
470p/50V



TP2
VGS_L



C42
4.7u/25V



C43
4.7u/25V



C44
100n/50V



C45
4.7u/25V



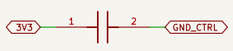
C46
100n/50V



C47
1u/50V



C48
100n/50V



R42
2.2k



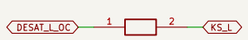
D5
US1M 1000V



R43
4.99k



R44
1k



C49
680p/50V



D6
BAT42W Schottky



R45
4.7k



R46
4.7k



L3
744226S



C50
10u/25V



C51
10u/25V



C52
10u/25V



C53
100n/50V



AI-GENERATED ENGINEERING PROTOTYPE — V2 — HUMAN REVIEW REQUIRED

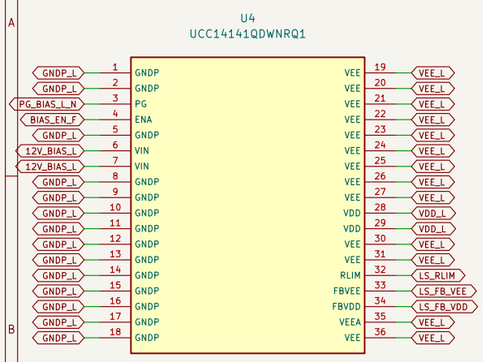
Sheet: /Low-side isolated gate driver and bias/
File: 03_gate_driver_ls.kicad_sch

Title: Low-side isolated gate driver and bias

Size: A3	Date:	**Rev: V2**
KiCad E.D.A. 10.0.7		Id: 6/12

Low-side isolated gate driver and bias (2)

Mirror of the high-side channel: IN+ = PWM_L_F, IN- = PWM_H_F; DESAT diode cathode on SW; KS_L is never joined to PGND on the PCB (KR-PCB-003). Isolation slot under the library SOIC-16W footprint is a layout task.



























AI-GENERATED ENGINEERING PROTOTYPE — V2 — HUMAN REVIEW REQUIRED

Sheet: /Low-side isolated gate driver and bias (2)/
File: 03_gate_driver_ls_2.kicad_sch

Title: Low-side isolated gate driver and bias (2)

Size: A3	Date:	**Rev: V2**
KiCad E.D.A. 10.0.7		Id: 7/12

Isolated voltage sensing island and its bias module

Third UCC14141 (CR-12): VDD-VEE 15 V (80.6 k / 16.2 k, A-P2), COM-VEE 5.0 V = 5V_ISO, VEE = PGND. AMC1311 ladders 4 x 750 k (VIN, 602 V full scale) and 2 x 750 k (VOUT, 302 V) with 10.0 k bottoms; the taps VIN_DIV_T1..T3 and VOUT_DIV_T1 are HV nodes (classes HV_VIN / HV_VOUT, 120 V steps).

L4
744226S
12V_PROT 1 W1A
GND_CTRL 2 W2A
W2B 3 GNDP_S
W1B 4 12V_BIAS_S

C60
10u/25V
12V_BIAS_S 1 2 GNDP_S

C61
10u/25V
12V_BIAS_S 1 2 GNDP_S

C62
10u/25V
12V_BIAS_S 1 2 GNDP_S

C63
100n/50V
12V_BIAS_S 1 2 GNDP_S

U5
UCC14141QDWNRQ1
GNDP_S 1 GNDP
GNDP_S 2 GNDP
PG_BIAS_S_N 3 PG
BIAS_EN_F 4 ENA
GNDP_S 5 GNDP
12V_BIAS_S 6 VIN
12V_BIAS_S 7 VIN
GNDP_S 8 GNDP
GNDP_S 9 GNDP
GNDP_S 10 GNDP
GNDP_S 11 GNDP
GNDP_S 12 GNDP
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GNDP_S 15 GNDP
GNDP_S 16 GNDP
GNDP_S 17 GNDP
GNDP_S 18 GNDP
VEE 19 PGND
VEE 20 PGND
VEE 21 PGND
VEE 22 PGND
VEE 23 PGND
VEE 24 PGND
VEE 25 PGND
VEE 26 PGND
VEE 27 PGND
VDD 28 ISO_15V
VDD 29 ISO_15V
VEE 30 PGND
VEE 31 PGND
RLIM 32 ISO_RLIM
FBVEE 33 ISO_FB_VEE
FBVDD 34 ISO_FB_VDD
VEEA 35 PGND
VEE 36 PGND

R53
80.6k
ISO_15V 1 2 ISO_FB_VDD

R54
16.2k
ISO_FB_VDD 1 2 PGND

C64
330p/50V
ISO_FB_VDD 1 2 PGND

R55
16.2k
5V_ISO 1 2 ISO_FB_VEE

R56
16.2k
ISO_FB_VEE 1 2 PGND

C65
330p/50V
ISO_FB_VEE 1 2 PGND

R57
1k
ISO_RLIM 1 2 5V_ISO

C66
4.7u/25V
ISO_15V 1 2 PGND

C67
4.7u/25V
ISO_15V 1 2 PGND

C68
100n/50V
ISO_15V 1 2 PGND

C69
4.7u/25V
5V_ISO 1 2 PGND

R58
10k
3V3 1 2 PG_BIAS_S_N

R59
750k
VIN 1 2 VIN_DIV_T1

R60
750k
VIN_DIV_T1 1 2 VIN_DIV_T2

R61
750k
VIN_DIV_T2 1 2 VIN_DIV_T3

R62
750k
VIN_DIV_T3 1 2 VIN_DIV

R63
10k
VIN_DIV 1 2 PGND

C70
1n/50V
VIN_DIV 1 2 PGND

C71
100p/50V
VIN_DIV 1 2 PGND

U6
AMC1311BDWVR
5V_ISO 1 VDD1
VIN_DIV 2 IN
PGND 3 SHTDN
PGND 4 GND1
GND2 5 GND_CTRL
OUTN 6 VIN_OUTN
OUTP 7 VIN_OUTP
VDD2 8 3V3

C72
100n/50V
5V_ISO 1 2 PGND

C73
100n/50V
3V3 1 2 GND_CTRL

R64
750k
VOUT 1 2 VOUT_DIV_T1

R65
750k
VOUT_DIV_T1 1 2 VOUT_DIV

R66
10k
VOUT_DIV 1 2 PGND

C74
1n/50V
VOUT_DIV 1 2 PGND

C75
100p/50V
VOUT_DIV 1 2 PGND

AI-GENERATED ENGINEERING PROTOTYPE — V2 — HUMAN REVIEW REQUIRED

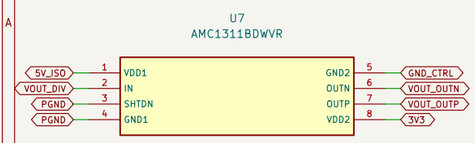
Sheet: /Isolated voltage sensing island and its bias module/
File: 04_sensing_island.kicad_sch

Title: Isolated voltage sensing island and its bias module

Size: A3	Date:	**Rev: V2**
KiCad E.D.A. 10.0.7		Id: 8/12

Isolated voltage sensing island and its bias module (2)

Third UCC14141 (CR-12): VDD-VEE 15 V (80.6 k / 16.2 k, A-P2), COM-VEE 5.0 V = 5V_ISO, VEE = PGND. AMC1311 ladders 4 x 750 k (VIN, 602 V full scale) and 2 x 750 k (VOUT, 302 V) with 10.0 k bottoms; the taps VIN_DIV_T1..T3 and VOUT_DIV_T1 are HV nodes (classes HV_VIN / HV_VOUT, 120 V steps).







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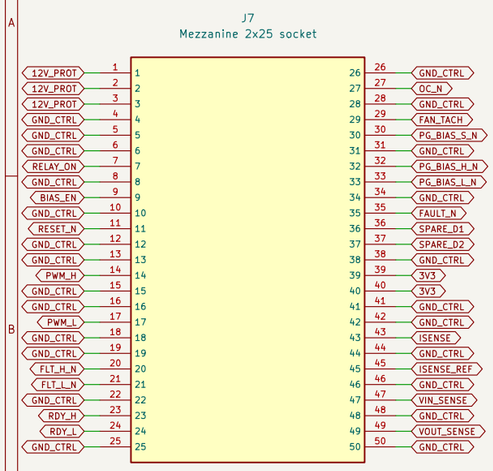
Sheet: /Isolated voltage sensing island and its bias module (2)/
File: 04_sensing_island_2.kicad_sch

Title: Isolated voltage sensing island and its bias module (2)

Size: A3	Date:	**Rev: V2**
KiCad E.D.A. 10.0.7		Id: 9/12

SELV strip: mezzanine, conditioning, relay and fan interface

Mezzanine socket SSW-125-01-G-D with the pin assignment of design_decisions.json; receiving networks 100 ohm / 100 pF / 10 k pull-down on RESET_N and BIAS_EN (A-P6); OPA2320 difference amplifiers gain 1.5; REF3125 -> VOC_SET 0.750 V (I_OC 37.5 A); ISENSE_REF = 0.1 x 3V3 (A-P1); AO3400A + SS14 relay coil drive; fan header with tacho pull-up. All nets on this sheet are SELV (CTRL_SELV).
PWR_FLAG symbols state supply provenance for ERC; they do not prove voltage, current capacity or protection.





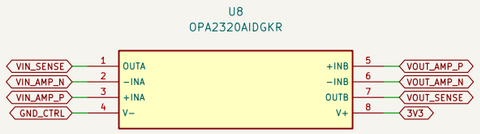




























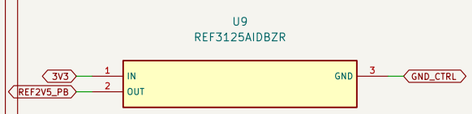


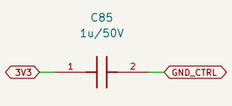


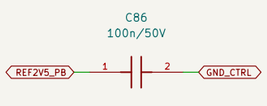
















AI-GENERATED ENGINEERING PROTOTYPE — V2 — HUMAN REVIEW REQUIRED

Sheet: /SELV strip: mezzanine, conditioning, relay and fan interface/
File: 05_selv_strip.kicad_sch

Title: SELV strip: mezzanine, conditioning, relay and fan interface

Size: A3	Date:	**Rev: V2**
KiCad E.D.A. 10.0.7		Id: 10/12

SELV strip: mezzanine, conditioning, relay and fan interface (2)

Mezzanine socket SSW-125-01-G-D with the pin assignment of design_decisions.json: receiving networks 100 ohm / 100 pF / 10 k pull-down on RESET_N and BIAS_EN (A-P6); OPA2320 difference amplifiers gain 1.5; REF3125 -> VOC_SET 0.750 V (I_OC 37.5 A); ISENSE_REF = 0.1 x 3V3 (A-P1); AO3400A + SS14 relay coil drive; fan header with tacho pull-up. All nets on this sheet are SELV (CTRL_SELV).
PWR_FLAG symbols state supply provenance for ERC; they do not prove voltage, current capacity or protection.

R81
10k
3V3
OC_N

C87
100n/50V
3V3
GND_CTRL

C88
1u/50V
3V3
GND_CTRL

R82
9.09k
3V3
ISENSE_REF

R83
1k
ISENSE_REF
GND_CTRL

C89
100n/50V
ISENSE_REF
GND_CTRL

J8
FAN 12V/GND/TACH
12V_PROT
GND_CTRL
FAN_TACH

R84
10k
3V3
FAN_TACH

J9
Relay coil K1
12V_PROT
K1_LOW

Q3
AO3400A
K1_LOW
RLY_G
GND_CTRL

R85
100
RELAY_ON
RLY_G

R86
10k
RLY_G
GND_CTRL

D7
SS14 Schottky
12V_PROT
K1_LOW

C90
100u/25V
12V_PROT
GND_CTRL

C91
10u/25V
12V_PROT
GND_CTRL

#FLG1
VIN_RAW
PWR_FLAG

#FLG2
VIN
PWR_FLAG

#FLG3
PGND
PWR_FLAG

#FLG4
12V_PROT
PWR_FLAG

#FLG5
3V3
PWR_FLAG

#FLG6
GND_CTRL
PWR_FLAG

#FLG7
KS_H
PWR_FLAG

#FLG8
KS_L
PWR_FLAG

#FLG9
5V_ISO
PWR_FLAG

#FLG10
12V_BIAS_H
PWR_FLAG

#FLG11
12V_BIAS_L
PWR_FLAG

#FLG12
12V_BIAS_S
PWR_FLAG

#FLG13
GNDP_H
PWR_FLAG

#FLG14
GNDP_L
PWR_FLAG

#FLG15
GNDP_S
PWR_FLAG

AI-GENERATED ENGINEERING PROTOTYPE — V2 — HUMAN REVIEW REQUIRED

Sheet: /SELV strip: mezzanine, conditioning, relay and fan interface (2)/
File: 05_selv_strip_2.kicad_sch

Title: SELV strip: mezzanine, conditioning, relay and fan interface (2)

Size: A3	Date:	**Rev: V2**
KiCad E.D.A. 10.0.7		Id: 11/12

Heatsink envelope, terminals, mounting, test points

Heatsink envelope LAM 5 D K 100 12 (corner holes are placeholders), 6 x M4 baseplate standoffs, 4 x M3 control-board standoffs, bare-PCB test pads. The PE stud and PGND-PE network are on sheet 01.

HS1
LAM 5 D K 100 12

H2
M4 standoff

H3
M4 standoff

H4
M4 standoff

H5
M4 standoff

H6
M4 standoff

H7
M4 standoff

H8
M3 standoff

H9
M3 standoff

H10
M3 standoff

H11
M3 standoff

TP3
3V3

3V3 1

TP4
GND_CTRL

GND_CTRL 1

TP5
GND_CTRL

GND_CTRL 1

TP6
PWM_H_F

PWM_H_F 1

TP7
PWM_L_F

PWM_L_F 1

TP8
FAULT_N

FAULT_N 1

TP9
ISENSE

ISENSE 1

TP10
VIN_SENSE

VIN_SENSE 1

TP11
VOUT_SENSE

VOUT_SENSE 1

TP12
SPARE_D1

SPARE_D1 1

TP13
SPARE_D2

SPARE_D2 1

AI-GENERATED ENGINEERING PROTOTYPE — V2 — HUMAN REVIEW REQUIRED

Sheet: /Heatsink envelope, terminals, mounting, test points/
File: 06_mechanical.kicad_sch

Title: Heatsink envelope, terminals, mounting, test points

Size: A3	Date:	**Rev: V2**
KiCad E.D.A. 10.0.7		Id: 12/12